

# Smart Loan Risk & Approval System

## Business Report

End-to-End Machine Learning Application  
for Loan Risk Prediction and Approval

Report Summary

Report Type:

Business Report

Generated On:

2026-09-28 20:20

Total Records:

5,000

Total Columns:

21

Target Column:

Loan\_Status

Active Model:

Logistic Regression

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### 1. Dataset Overview

|               |               |              |                  |
|---------------|---------------|--------------|------------------|
| Total Records | Total Columns | Numeric Cols | Categorical Cols |
| 5,000         | 21            | 11           | 10               |

### 2. Data Quality

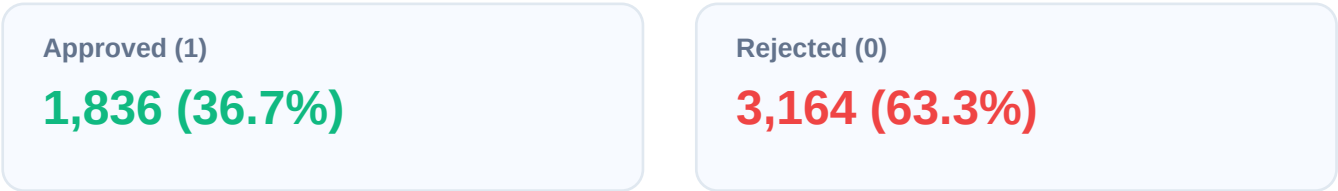
| Metric           | Count | Percentage |
|------------------|-------|------------|
| Missing Values   | 0     | 0.00%      |
| Duplicate Rows   | 0     | 0.00%      |
| Numerical Cols   | 11    | 52%        |
| Categorical Cols | 10    | 48%        |

Dataset is ready for analysis and model training.

### 3. Missing Values by Column

No missing values detected.

### 4. Target Analysis (Loan\_Status)



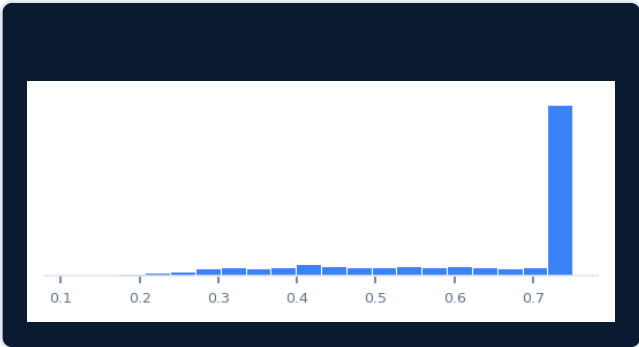
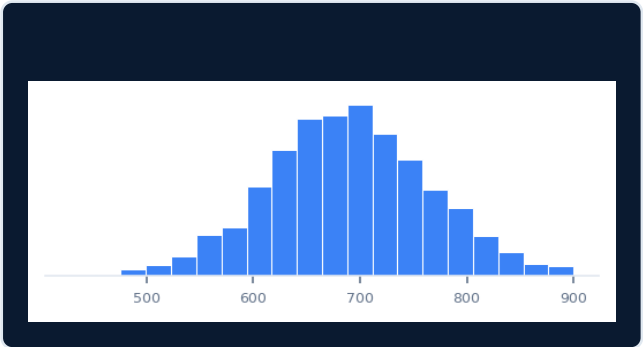
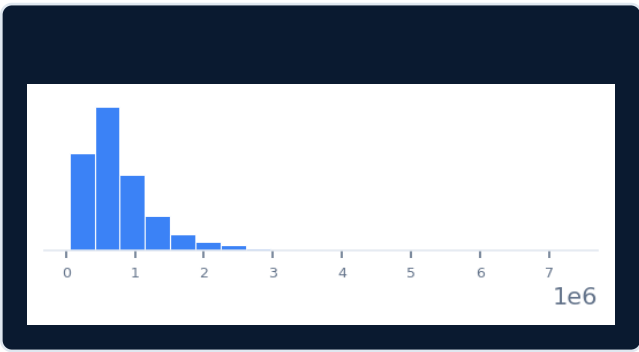
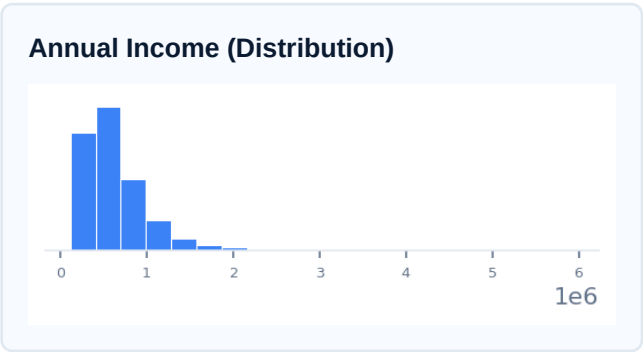
### 5. Descriptive Statistics

| Feature            | Mean      | Median    | Min      | Max        | Std Dev   |
|--------------------|-----------|-----------|----------|------------|-----------|
| Age                | 42.9      | 43.0      | 21.0     | 65.0       | 13.0      |
| Employment_Years   | 15.7      | 15.1      | -0.0     | 35.0       | 9.1       |
| Annual_Income      | 630699.8  | 540000.0  | 120000.0 | 5942000.0  | 375636.2  |
| Coapplicant_Income | 102987.6  | 65000.0   | 0.0      | 1856000.0  | 140083.2  |
| Loan_Amount        | 792643.4  | 638000.0  | 51000.0  | 7344000.0  | 588091.3  |
| Loan_Term_Years    | 8.8       | 7.0       | 2.0      | 20.0       | 5.3       |
| Cibil_Score        | 689.4     | 689.0     | 429.0    | 900.0      | 75.4      |
| Existing_Loans     | 1.2       | 1.0       | 0.0      | 6.0        | 1.1       |
| DTI_Ratio          | 0.6       | 0.8       | 0.1      | 0.8        | 0.2       |
| Residential_Assets | 2080798.6 | 1369500.0 | 66000.0  | 80846000.0 | 2521706.4 |
| Bank_Assets        | 478127.0  | 343000.0  | 22000.0  | 7727000.0  | 458559.1  |

## 6. Outlier Analysis (IQR Method)

| Feature            | Outlier Count | Outlier % |
|--------------------|---------------|-----------|
| Annual_Income      | 227           | 4.5%      |
| Coapplicant_Income | 228           | 4.6%      |
| Loan_Amount        | 282           | 5.6%      |
| Cibil_Score        | 43            | 0.9%      |
| Residential_Assets | 372           | 7.4%      |
| Bank_Assets        | 309           | 6.2%      |

## 7. Distribution of Key Features



## 8. Model Training & Comparison

| Model               | Accuracy | Precision | Recall | F1 Score | ROC-AUC |
|---------------------|----------|-----------|--------|----------|---------|
| Logistic Regression | 0.767    | 0.786     | 0.854  | 0.819    | 0.841   |

Only one successfully evaluated model is available.

Selected Model: Logistic Regression

### Model Performance Summary

The Logistic Regression model was selected based on optimal evaluation metrics. It achieved an accuracy of 76.7% on the validation set, demonstrating strong generalization capability for predicting loan risk.

## 9. Active Model Details

Model Name

Logistic Regression

Model Version

v20260928201830

Model Type

Loan Risk Classification

Training Status

Ready for Prediction

### 10. New Applicant Prediction

Applicant Information

No applicant data available.

Prediction Result

Prediction:  
**LOW RISK**

Model Confidence: 79.4%

Approval Probability: 79.4%

Risk Probability: 20.6%

Decision: **APPROVE**

### 11. Final Summary

The system analyzed 5,000 historical loan applications using Logistic Regression. The selected model achieved 76.7% accuracy on the evaluation dataset. The latest applicant was classified as LOW RISK with 79.4% confidence.